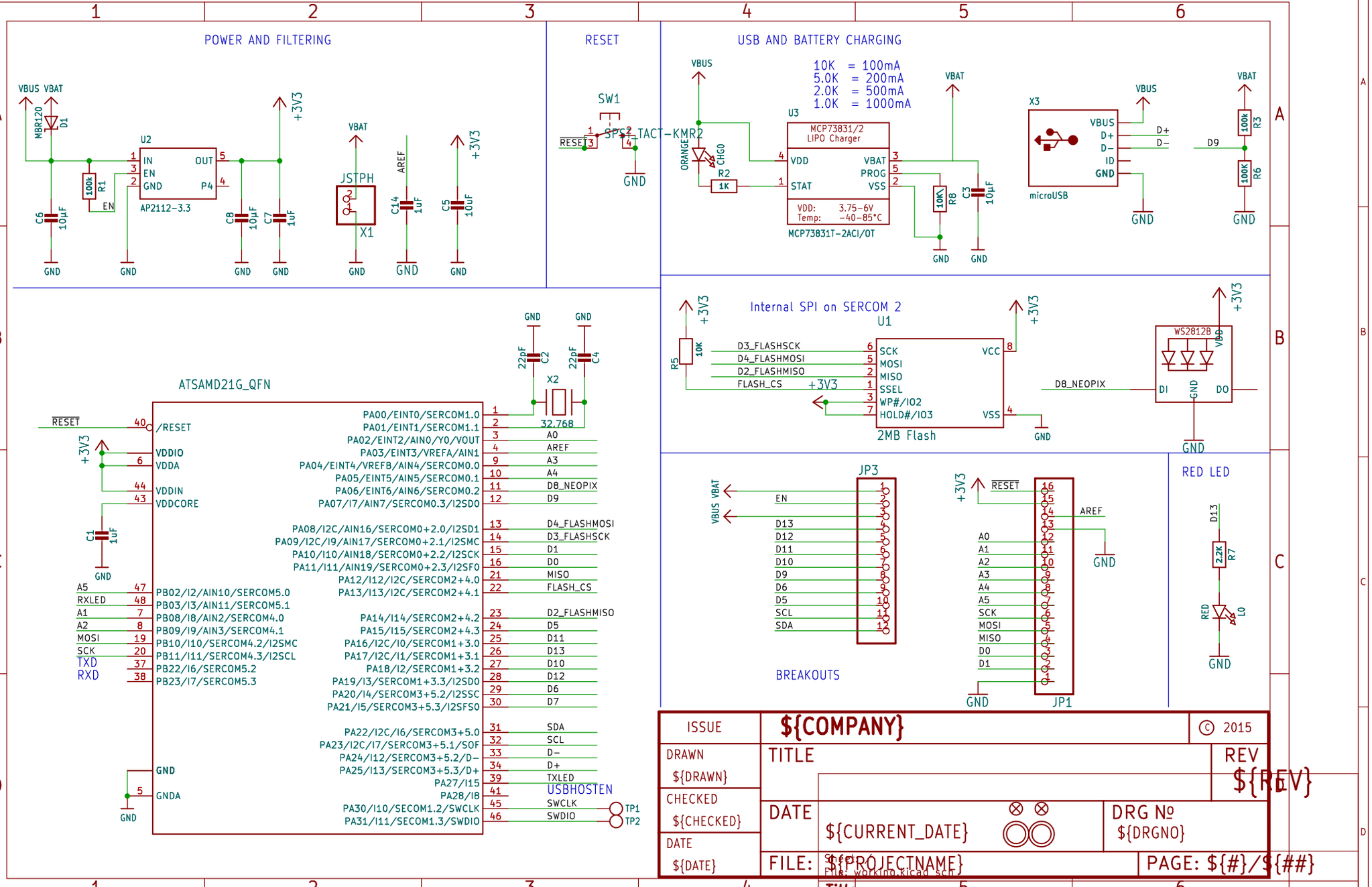
POWER AND FILTERING
VBUS VBAT
MBR120
D1
U2
IN
EN
GND
OUT
P4
AP2112-3.3
100k
R1
EN
C6
10µF
C8
10µF
C7
1uF
+3V3
JSTPH
X1
VBAT
AREF
C14
1uF
C5
10uF
RESET
SW1
SPST-TACT-KMR2
RESET
GND
USB AND BATTERY CHARGING
VBUS
10K = 100mA
5.0K = 200mA
2.0K = 500mA
1.0K = 1000mA
ORANGE
CHGO
R2
1K
U3
MCP73831/2
LIPO Charger
VDD
STAT
VBAT
PROG
VSS
VDD: 3.75-6V
Temp: -40-85°C
MCP73831T-2ACI/OT
10K
R8
C3
10µF
X3
microUSB
VBUS
D+
D-
ID
GND
D9
100K
R3
100K
R6
ATSAMD21G_QFN
22pF
C2
22pF
C4
X2
32.768
RESET
/RESET
VDDIO
VDDA
VDDIN
VDDCORE
C1
1uF
A5
RXLED
A1
A2
MOSI
SCK
TXD
RXD
PB02/I2/AIN10/SERCOM5.0
PB03/I3/AIN11/SERCOM5.1
PB08/I8/AIN2/SERCOM4.0
PB09/I9/AIN3/SERCOM4.1
PB10/I10/SERCOM4.2/I2SMC
PB11/I11/SERCOM4.3/I2SCL
PB22/I6/SERCOM5.2
PB23/I7/SERCOM5.3
GNDA
PA00/EINT0/SERCOM1.0
PA01/EINT1/SERCOM1.1
PA02/EINT2/AIN0/Y0/VOUT
PA03/EINT3/VREFA/AIN1
PA04/EINT4/VREFB/AIN4/SERCOM0.0
PA05/EINT5/AIN5/SERCOM0.1
PA06/EINT6/AIN6/SERCOM0.2
PA07/I7/AIN7/SERCOM0.3/I2SD0
PA08/I2C/AIN16/SERCOM0+2.0/I2SD1
PA09/I2C/I9/AIN17/SERCOM0+2.1/I2SMC
PA10/I10/AIN18/SERCOM0+2.2/I2SCK
PA11/I11/AIN19/SERCOM0+2.3/I2SF0
PA12/I12/I2C/SERCOM2+4.0
PA13/I13/I2C/SERCOM2+4.1
PA14/I14/SERCOM2+4.2
PA15/I15/SERCOM2+4.3
PA16/I2C/I0/SERCOM1+3.0
PA17/I2C/I1/SERCOM1+3.1
PA18/I2/SERCOM1+3.2
PA19/I3/SERCOM1+3.3/I2SD0
PA20/I4/SERCOM3+5.2/I2SSC
PA21/I5/SERCOM3+5.3/I2SFS0
PA22/I2C/I6/SERCOM3+5.0
PA23/I2C/I7/SERCOM3+5.1/SOF
PA24/I12/SERCOM3+5.2/D-
PA25/I13/SERCOM3+5.3/D+
PA27/I15
PA28/I8
PA30/I10/SECOM1.2/SWCLK
PA31/I11/SECOM1.3/SWDIO
A0
AREF
A3
A4
D8_NEOPIX
D9
D4_FLASHMOSI
D3_FLASHSCK
D1
D0
MISO
FLASH_CS
D2_FLASHMISO
D5
D11
D13
D10
D12
D6
D7
SDA
SCL
D-
D+
TXLED
USBHOSTEN
SWCLK
SWDIO
TP1
TP2
Internal SPI on SERCOM 2
U1
R5
10K
D3_FLASHSCK
D4_FLASHMOSI
D2_FLASHMISO
FLASH_CS
SCK
MOSI
MISO
SSEL
WP#/IO2
HOLD#/IO3
VCC
VSS
2MB Flash
WS2812B
DI
DO
D8_NEOPIX
JP3
VBUS VBAT
EN
D13
D12
D11
D10
D9
D6
D5
SCL
SDA
BREAKOUTS
JP1
RESET
AREF
A0
A1
A2
A3
A4
A5
SCK
MOSI
MISO
D0
D1
RED LED
D13
2.2K
R7
RED
L0
ISSUE
${COMPANY}
© 2015
DRAWN
${DRAWN}
TITLE
REV
${REV}
CHECKED
${CHECKED}
DATE
${DATE}
${CURRENT_DATE}
DRG №
${DRGNO}
FILE: ${PROJECTNAME}
PAGE: ${#}/${##}
Title:
Size: User
Date:
Rev:
KiCad E.D.A. kicad-cli 7.0.6-7.0.6-ubuntu22.04.1
Id: 1/1